Field	This example
Team name	**Viounique**
Project title	**Assistance application for Disasters**
SDLC chosen in M1 + reason	**We selected the Agile SDLC for its flexibility to handle evolving requirements and support continuous future development. Priorities are ordered each sprint (MoSCoW) while the Must set stays stable.**
Date / version	**September 4, 2026**
GitHub repo	https://github.com/Viounique

1. Introduction

1.1 Purpose and scope

In 2025, Myanmar was struck by catastrophic flooding driven by intense monsoon rains and regional cyclones. Low-lying river basins and coastal regions were submerged, leaving tens of thousands of residents trapped, critical infrastructure damaged, and communication networks severed. Rescue agencies struggled to locate victims in urgent need and distribute life-saving supplies efficiently.

In 2025–2026, Thailand faced severe, recurring floods across both northern river basins and southern urban hubs. Densely populated communities and remote mountain villages were suddenly cut off by flash floods and mudslides. Emergency responders were overwhelmed by fragmented, delayed reports, making it nearly impossible to prioritize vulnerable populations such as the bedridden, elderly, and infants.

These regional disasters reveal a critical, shared vulnerability: the lack of a unified, intelligent system for emergency reporting and crisis coordination. Victims are unable to send real-time location data or specify special care needs, coordinators lack live operational visibility, and rescue teams are forced to deploy blindly. Precious hours are lost, and lives are endangered simply because critical information fails to flow when it is needed most.

Purpose of this document. To specify what the Emergency Assistance System must do — first for the MVP (Must requirements), then for later iterations — in a form that can be designed, prototyped and tested. In one sentence: the system converts victims' distress requests into a prioritised, real-time operational map for coordinators and rescue teams.

MVP success measures (every one is numeric and checked in §3–§4):

- A first-time victim submits a complete request (GPS + emergency type) in ≤ 45 s (NFR-3).
- A request made offline reaches the server within 60 s of reconnection, with 0 requests lost (AC-2.2, NFR-4).
- A new request appears on the coordinator's map within 10 s (NFR-7), and 100% of requests with a care flag are scored Red (AC-4.1).
- A coordinator can verify a case by phone and record its status from the same screen (AC-5.1, AC-5.2).

Problem register — the numbered pains that every requirement traces back to:

ID	Pain observed	Who feels it	Source	Addressed by

ID	Pain observed	Who feels it	Source	Addressed by
P1	Victims cannot describe their exact location by phone, or lose signal before finishing complex forms; rescuers miss their whereabouts.	Victim	§1.1; stakeholder table	FR-1
P2	Unstable networks make online forms fail silently; victims never learn that their call for help was lost.	Victim	§1.1; stakeholder table	FR-2
P3	Coordinators get fragmented reports from many channels with no spatial context; affected clusters are hard to see.	Coordinator	§1.1; stakeholder table	FR-3
P4	High call volume buries life-threatening cases; bedridden, infant and elderly victims are not prioritised.	Coordinator, Victim	§1.1	FR-4
P5	Rescue teams waste time travelling to victims who have evacuated or whose details are false, wrong or duplicated.	Rescue team, Coordinator	Stakeholder table (fears)	FR-5 (+ FR-1 duplicate control)
P6	Victims fear delay and cannot tell whether anyone received their request.	Victim	Stakeholder table (fears)	FR-6
P7	Rescue teams “deploy blindly” — no priority or vulnerable-group details for their targets.	Field rescue team	§1.1; stakeholder table	FR-7
P8	A full form is slow and error-prone in a crisis with weak signal (derived from P1).	Victim	Derived from P1	FR-8
P9	Agencies and NGOs need transparent, standardised rescue logs but fear personal-data leaks.	Agencies, NGOs	Stakeholder table	FR-9

1.2 Users and stakeholders

Name / role	Kind	What they need	What they fear / forbid
Flood Victim / Affected Citizen (primary user)	primary user	Simple way to send GPS coordinates and request special medical/food aid without complex setup. (FR-1, FR-2, FR-6, FR-8)	Losing signal and requests failing silently; delay in critical rescue teams reaching them.
Field Rescue Team / First Responder (primary user)	primary user	Clear priority triage (colour codes) and exact locations of vulnerable groups (e.g., bedridden, infants). (FR-4, FR-5 via dispatch; own view in FR-7)	Wasting resources on fake/duplicate reports; entering dangerous flood zones without proper equipment.
Emergency Rescue Coordinator / Operations Center	Secondary stakeholder	Real-time live map visibility, centralized data from local/national agencies, and efficient resource allocation. (FR-3, FR-4, FR-5, FR-7, FR-9)	System overload or crashes during peak disaster hours; fragmented, conflicting information leading to missed cases.
Government Agencies & NGOs (e.g., Thai DDPM, WFP, UNICEF)	Institution / Partner	Standardized data integration, transparent rescue logs, and secure access to demographic/needs data. (FR-9, NFR-5, NFR-6, NFR-11)	Data privacy breaches; personal health or identity data leaking into unverified public channels.

The field rescue team is a primary user whose own screen is a Should (FR-7). In the MVP they are served through the coordinator, who verifies and dispatches (FR-4, FR-5) — a deliberate trade-off to keep the MVP small.

2. Overall description

2.1 Product context — what it does / does not

Emergency Assistance System converts distress requests into a real-time operational map for coordinators and rescue teams. It has a victim mobile client that works offline, a coordinator web dashboard (map, queue, case actions) and a server that stores requests, scores priority and pushes updates.

IN SCOPE (MVP this term)	OUT OF SCOPE (explicit promise)
• FR-1 Victim Request Logging: Log distress requests via app from mobile (GPS, type, care flags). • FR-2 Offline Capture & Sync: Capture requests offline and sync when online. • FR-3 Rescue Coordinator Live Map: Display requests on a real-time map with filters. • FR-4 Priority Scoring (Rule-based): Rule-based Red/Yellow/Green scoring to prioritize requests quickly. • FR-5 Contact Victim & Case Status: Call the victim and record the case status.	• No group chat or direct text messaging inside the app. • No payment processing or donation subscription feature. • No AI tutor or medical assistant to answer clinical questions. • No reading or summarizing external LINE/Messenger groups. • No accident / vehicle-breakdown reports; no live integration with external agency systems; no AI-based priority scoring (see §6).

MoSCoW beyond the MVP. Should (built if time allows): FR-6 Victim Status Notification; FR-7 Assign Rescue Team & Field Case View. Could: FR-8 AI Suggestion from Free Text; FR-9 Operation Report Export. **Must = MVP = FR-1 … FR-5.** Won't this term: see §6.

2.2 Assumptions and constraints

- **Assumption:** Every victim or volunteer has a smartphone capable of capturing GPS coordinates and connecting to cellular or internet networks, at least intermittently.
- **Assumption:** Coordinators work at an operations centre with a stable internet connection and a desktop browser; their sectors are assigned by an administrator before use (sector set-up itself is out of scope — NFR-5 relies on it).
- **Assumption:** Cellular voice calls work often enough for verification calls (FR-5).
- **Constraint:** One LLM API call per natural-language request parsing (FR-8 only) — no custom model training, no fine-tuning.
- **Constraint:** Thai and English UI text; no paid API beyond the single LLM call above.
- **Constraint:** Personal data — we store victim name, phone number, GPS location, emergency type, care flags, number of people and status history only; we do not store message content from any chat app.
- **Constraint:** Course schedule: SRS + Design due at M2; a clickable prototype (no real backend/DB required) at M3; all work is committed to the team GitHub repo.
- **Constraint:** Agile: the Must set (FR-1 … FR-5) is frozen for the MVP; Should/Could may be re-ordered each sprint.

3. Functional requirements

Format: As a <role>, I want <goal>, so that <benefit>. Every story below traces to a pain (P1–P9, §1.1) observed in a (simulated) stakeholder scenario, and every acceptance criterion is something a test could actually check. Each FR has at least one [Happy] and at least one [Edge] criterion.

3.1 Requirement summary and priority (MoSCoW)

ID	Requirement	MoSCoW	Actor	Pain	Use case	Size*
FR-1	Victim Request Logging	Must · MVP	Victim	P1	UC-1	M
FR-2	Offline Capture & Automatic Sync	Must · MVP	Victim	P2	UC-2	M
FR-3	Coordinator Live Map with Filters	Must · MVP	Coordinator	P3	UC-3, UC-4	L
FR-4	Rule-based Priority Scoring	Must · MVP	System / Coordinator	P4	UC-5	S
FR-5	Contact Victim & Case Status	Must · MVP	Coordinator	P5	UC-6, UC-7	M
FR-6	Victim Status Notification	Should	Victim	P6	UC-8	M
FR-7	Assign Rescue Team & Field Case View	Should	Coordinator, Field Rescue Team	P7	UC-9, UC-12	M

ID	Requirement	MoSCoW	Actor	Pain	Use case	Size*
FR-8	AI Suggestion from Free Text	Could	Victim, AI Module	P8	UC-10	M
FR-9	Operation Report Export	Could	Coordinator	P9	UC-11	S

* Initial relative size estimate (S / M / L) by the team, to be refined at sprint planning.

3.2 Detailed requirements and acceptance criteria

FR-1 — Victim Request Logging [Must · MVP]

User story: As a flood victim, I want to submit a distress request with my GPS location, emergency type and any special-care needs, so that rescue teams can find me and bring the right help without a long phone call.

Pain this traces to: *Victims struggle to describe their location by phone, or lose signal before finishing a complex form, so rescue teams miss their exact whereabouts (P1).*

Data captured (within constraint on personal data): *name, phone, GPS latitude/longitude (+ accuracy), emergency type (Medical / Evacuation / Food & Water / Other), care flags (bedridden, infant, elderly, none), number of people.*

1. ***[Happy]*** *Given GPS permission is granted and an emergency type is selected, when the victim taps Submit, then a request is stored with latitude, longitude, type, phone and timestamp, its status is "Submitted", and a reference ID is shown on screen.*
2. ***[Happy]*** *Given an active request, when the victim edits phone, type or care flags, then the same record is updated, its priority is re-scored, and no second record is created.*
3. ***[Edge]*** *Given GPS is denied or unavailable, when the victim submits, then the app asks them to pin their location on a map or type a landmark, accepts the request, and flags it "location unverified".*
4. ***[Edge]*** *Given the emergency type or phone number is empty, when the victim taps Submit, then no request is created and the missing field is highlighted with a message in the selected language (Thai/English).*
5. ***[Edge]*** *Given the victim already has an active request with the same phone number, when they submit a new one, then the app offers "Update existing request?" instead of creating a duplicate.*

FR-2 — Offline Capture & Automatic Sync [Must · MVP]

User story: As a flood victim, I want my request saved on my phone when there is no signal and sent automatically when the connection returns, so that my call for help is never silently lost.

Pain this traces to: *Unstable mobile networks in disaster zones make online forms fail silently; victims do not know their call for help was lost (P2).*

1. ***[Happy]*** *Given no internet connection, when the victim submits, then the request is saved on the device with status "Offline – pending" and a banner reads "Not sent yet – will send automatically".*
2. ***[Happy]*** *Given a pending request, when connectivity returns, then the app uploads it within 60 s, the status becomes "Submitted", and the server holds exactly one copy.*
3. ***[Edge]*** *Given a pending request, when the app is force-closed or the phone restarts, then on reopening the request is still pending and is synced as soon as a connection exists (NFR-4).*
4. ***[Edge]*** *Given an upload fails (timeout or server error), when the app retries every 30 s, then the status stays "Offline – pending" and every retry reuses the same client-generated request ID, so no duplicate is created.*

FR-3 — Coordinator Live Map with Filters [Must · MVP]

User story: As a rescue coordinator, I want to see all active requests on a real-time map with filters, so that I can see which areas are worst hit and allocate resources efficiently.

Pain this traces to: *Coordinators receive fragmented reports from many channels with no spatial context, so heavily affected clusters are hard to identify (P3).*

1. ***[Happy]*** *Given active requests exist in the coordinator's sector, when the coordinator opens the map, then every active request is plotted as a pin at its recorded coordinates, coloured Red / Yellow / Green by priority.*
2. ***[Happy]*** *Given pins are displayed, when the coordinator applies a filter (emergency type, priority,*

status or vulnerability flag), then only matching pins remain and the number of matches is shown.

3. ***[Happy]** Given the map is open, when a new request reaches the server, then its pin appears within 10 s without a manual refresh (NFR-7).*
4. ***[Edge]** Given there are no active requests, or none match the filter, when the map loads or the filter is applied, then an empty-state message ("No matching requests") and a Clear-filter button are shown — no error.*
5. ***[Edge]** Given a request flagged "location unverified", when it is displayed, then its pin has a dashed outline and the tooltip "Location not confirmed".*

FR-4 — Rule-based Priority Scoring [Must · MVP]

User story: As a rescue coordinator, I want every request to get an automatic Red / Yellow / Green priority from fixed safety rules, so that life-threatening and vulnerable cases sit at the top of my queue instead of being buried.

***Pain this traces to:** High volumes of calls overwhelm rescue centres and bury life-threatening cases (bedridden, infants, elderly) under routine requests (P4).*

***Scoring rules** (single source of truth for AC-4.1 … 4.5; the highest severity that matches wins):*

Priority	Rule
Red	Any care flag is set (bedridden, infant, elderly) OR emergency type = Medical.
Yellow	Emergency type = Evacuation or Other with no care flag; OR data incomplete/contradictory (tagged "needs review").
Green	Emergency type = Food & Water only, with no care flag and complete data.

1. ***[Happy]** Given a request with the flag bedridden, infant or elderly, when it is submitted, then its priority is Red within 2 s (NFR-8).*
2. ***[Happy]** Given a request of type Food & Water with no care flags, when it is submitted, then its priority is Green.*
3. ***[Happy]** Given the case queue, when it is displayed, then it is sorted Red → Yellow → Green, oldest first within the same colour.*
4. ***[Edge]** Given a request that matches more than one rule (e.g. Food & Water + infant), when it is scored, then the highest severity applies (Red).*
5. ***[Edge]** Given a request with incomplete or contradictory data, when it is scored, then it is Yellow with a "needs review" tag — never Green.*

FR-5 — Contact Victim & Case Status [Must · MVP]

User story: As a rescue coordinator, I want to call the victim from the dashboard and record the case status, so that I can verify the situation before dispatch and no team is sent to a false or already-resolved case.

***Pain this traces to:** Rescue teams waste critical time travelling through flooded zones only to find the victim has evacuated or gave incorrect details (P5).*

***Case status flow:** Submitted → Verified → Dispatched → Rescued; side exits: Unreachable, Canceled.*

1. ***[Happy]** Given a request open on the dashboard, when the coordinator clicks Call, then the device dialer opens with the victim's stored phone number.*
2. ***[Happy]** Given a request, when the coordinator sets the status to Verified, Dispatched or Rescued, then the new status, time and coordinator name are saved and shown on the dashboard and map within 10 s (NFR-7).*
3. ***[Edge]** Given a request with a missing or invalid phone number, when the case is opened, then the Call button is disabled with the tooltip "No valid number".*
4. ***[Edge]** Given a false or unconfirmable request, when the coordinator sets "Unreachable" or "Canceled" and picks a reason, then the pin leaves the Active view, the change is reflected across the dashboard, and the case stays in the case log (never deleted).*

FR-6 — Victim Status Notification [Should]

User story: As a flood victim, I want to be notified when my request is verified or a team is dispatched, so that I know help is on the way and do not have to keep calling or leave a safe spot.

Pain this traces to: *Victims fear delay and uncertainty — they cannot tell whether anyone received their request (P6).*

1. ***[Happy]*** *Given an active request, when the coordinator sets status Verified or Dispatched, then the victim's app shows a notification within 30 s (NFR-2).*
2. ***[Happy]*** *Given the victim opens the app, when the status screen loads, then it shows the current status and the time it changed.*
3. ***[Edge]*** *Given the victim's phone is offline when the status changes, when it reconnects, then the notification is delivered once and the status screen shows the latest status.*
4. ***[Edge]*** *Given the coordinator saves the same status twice, when the second save happens, then no second notification is sent.*

FR-7 — Assign Rescue Team & Field Case View [Should]

User story: As a rescue coordinator, I want to assign a rescue team to a case, and as a field rescuer I want to see my assigned cases with priority, location and vulnerable-group details, so that teams go to the right place prepared with the right equipment.

Pain this traces to: *Rescue teams are forced to deploy blindly, without priority or vulnerable-group information for their targets (P7).*

1. ***[Happy]*** *Given a Verified request, when the coordinator selects an available team, then the status becomes "Dispatched" and the team name is shown on the case.*
2. ***[Happy]*** *Given a team has assignments, when a team member opens "My cases", then only their assigned cases appear, sorted by priority, each with location, care flags and a navigation link.*
3. ***[Edge]*** *Given a team already has an active Red case, when the coordinator assigns another case to it, then a warning "Team busy on Red case" is shown and confirmation is required.*
4. ***[Edge]*** *Given an assignment is cancelled or reassigned, when the change is saved, then the former team's list updates within 10 s and the case returns to "Verified" until reassigned.*

FR-8 — AI Suggestion from Free Text [Could]

User story: As a flood victim on a weak connection, I want to describe my situation in one sentence and have the app suggest the emergency type and special-care flags, so that I finish the request with fewer taps.

Pain this traces to: *Typing and selecting through a full form is slow and error-prone in a crisis; victims lose signal before finishing (P8, derived from P1).*

1. ***[Happy]*** *Given the victim types "my mother cannot walk and the water is rising", when they tap Suggest, then the form pre-selects Medical + bedridden and asks the victim to confirm before submitting.*
2. ***[Edge]*** *Given the LLM API is unavailable or does not answer within 5 s, when the victim taps Suggest, then the form falls back to manual selection with a short message and submission is never blocked.*
3. ***[Edge]*** *Given the AI suggestion differs from what the victim selects, when the victim submits, then the victim's selection is used, and priority is still computed only by the FR-4 rules.*

FR-9 — Operation Report Export [Could]

User story: As a rescue coordinator, I want to export a summary of requests by status, priority and time range, so that agencies and NGOs get transparent rescue logs without exposing personal data.

Pain this traces to: *Agencies and NGOs need standardised, transparent rescue logs but fear personal data leaking (P9).*

1. ***[Happy]*** *Given a date range, when the coordinator clicks Export, then a CSV downloads within 10 s for up to 1,000 requests, one row per request: ID, created time, type, priority, status, resolved time.*
2. ***[Edge]*** *Given a date range with no requests, when the coordinator clicks Export, then no file is created and "No data in this range" is shown.*
3. ***[Edge]*** *Given any export, when the file is opened, then it contains no victim name or phone number —*

request IDs only (NFR-11).

3.3 INVEST quality gate

Each story is checked against INVEST — Independent, Negotiable, Valuable, Estimable, Small, Testable. ✓ = passes; ✓* = passes with the noted mitigation.

FR	I	N	V	E	S	T	Note / mitigation
FR-1	✓	✓	✓	✓	✓	✓	Core entry point; five criteria, all checkable on one form.
FR-2	✓*	✓	✓	✓	✓	✓	I: builds on FR-1's submit action; developed and tested against a stub upload endpoint.
FR-3	✓	✓	✓	✓	✓*	✓	S: largest Must (L); build the map first, filters second. Tested with seeded data.
FR-4	✓	✓	✓	✓	✓	✓	A pure rule function — testable with a table of inputs and expected colours.
FR-5	✓*	✓	✓	✓	✓	✓	I: needs an existing case; tested with seeded cases. Call + status form one "verify" workflow.
FR-6	✓*	✓	✓	✓	✓	✓	I: triggered by an FR-5 status change; testable by calling the status API directly.
FR-7	✓	✓	✓	✓	✓*	✓	S: two screens; if capacity is short, split into 7a (assign) and 7b (field view).
FR-8	✓	✓	✓	✓*	✓	✓	E: LLM output varies; time-boxed and tested with 10 fixed sample sentences.
FR-9	✓	✓	✓	✓	✓	✓	Small read-only export of existing data.

4. Non-functional requirements

Forbidden words unless quantified: fast, easy, user-friendly, smart, secure. NFR-5 (security/privacy) is required for M2 — every SRS must have at least one.

ID	Kind	Statement (testable)	How we will check
NFR-1	Performance	The live map shall load and display up to 200 active markers within 3 s on a standard 4G connection. (traces to FR-3)	Browser network profiler on a test phone, 200 seeded requests; average of 3 trials ≤ 3 s.
NFR-2	Performance	A status-change notification shall reach the victim's phone within 30 s of the coordinator saving the status, in ≥ 95% of trials. (traces to FR-6)	Compare server timestamp with receipt timestamp on the device; 20 trials.
NFR-3	Usability	A first-time victim shall submit a complete request (GPS + emergency type) in ≤ 45 s on average, and ≥ 4 of 5 testers shall finish with no help. (traces to FR-1)	Timed trial with 5 users who have never seen the app; stopwatch + observer.
NFR-4	Reliability	Pending offline requests shall survive an app crash or phone restart with 0 requests lost (10 of 10 trials). (traces to FR-2)	Create a pending request, force-close the app and restart the phone, reconnect; verify it syncs; 10 trials.
NFR-5	Security / privacy (required)	A coordinator shall access only requests in their authorised sector; requesting another sector's request ID via the API shall return 403 Forbidden in 10 of 10 attempts, with 0 foreign records in the response. (traces to FR-3, FR-5, FR-7)	Alter the request-ID and sector-ID parameters in 10 API calls using a coordinator token; check status code and body.
NFR-6	Security	100% of client–server traffic shall use HTTPS with TLS 1.2 or higher; 0 plain-HTTP endpoints. (traces to All)	Automated scan of all endpoints (e.g. SSL Labs / testssl.sh) and a code search for "http://".

ID	Kind	Statement (testable)	How we will check
NFR-7	Timeliness	A new request or status change shall appear on every coordinator's map within 10 s of reaching the server, in ≥ 95% of trials. (traces to FR-3, FR-5)	Submit or change 20 cases with two dashboards open; compare timestamps.
NFR-8	Performance	The priority score shall be calculated within 2 s of the server receiving a request, for 100% of 20 test requests. (traces to FR-4)	Log receipt and score timestamps; automated test with 20 requests covering all rules.
NFR-9	Capacity	The system shall handle 500 simultaneous request submissions and 20 concurrent coordinators with an error rate ≤ 1% and 95th-percentile response time ≤ 3 s. (traces to FR-1–FR-3)	Load test (k6 or JMeter) for 5 minutes at the stated load.
NFR-10	Localisation / accessibility	100% of UI strings exist in Thai and English (0 missing keys); touch targets ≥ 48 × 48 px; text contrast ≥ 4.5 : 1; Lighthouse accessibility score ≥ 90. (traces to FR-1, FR-3, FR-4)	String-key diff script; Lighthouse audit on the victim form and coordinator map.
NFR-11	Privacy (data minimisation)	The system shall store only the 7 approved personal-data fields (name, phone, GPS, emergency type, care flags, people count, status history); 0 additional personal fields; exports contain 0 names or phone numbers. (traces to FR-1, FR-9)	Review the database schema against the approved list; open a sample export and search for names/phones.

5. Use cases

5.1 Use-case list

Use case	Actor	Related FR	Brief success story
UC-1 Log Distress Request	Victim	FR-1	Victim enters emergency type and care needs; GPS is attached; the request appears in the coordinator queue with a reference ID.
UC-2 Sync Offline Requests	Victim app → Server	FR-2	App detects restored connection and uploads pending requests; status changes from "Offline – pending" to "Submitted".
UC-3 View Live Map	Coordinator	FR-3	Coordinator opens the map and sees all active requests of their sector as colour-coded pins.
UC-4 Filter Requests	Coordinator	FR-3	Coordinator filters by type, priority, status or care flag to isolate a group of cases.
UC-5 Score Priority (rules)	System (rule engine)	FR-4	On every new or edited request the system assigns Red / Yellow / Green and re-sorts the queue.
UC-6 Contact Victim	Coordinator	FR-5	Coordinator taps Call to verify the victim's condition before dispatching help.
UC-7 Update Case Status	Coordinator	FR-5	Coordinator records Verified / Dispatched / Rescued / Unreachable / Canceled; all dashboards update.
UC-8 Receive Status Alert	Victim	FR-6	Victim receives a notification when the case is verified or a team is dispatched.
UC-9 Assign Rescue Team	Coordinator	FR-7	Coordinator assigns an available team to a Verified, high-priority case.
UC-10 Suggest Type from Free Text (AI)	Victim, AI Module	FR-8	AI proposes emergency type and care flags from one sentence; victim confirms.
UC-11 Export Operation Report	Coordinator	FR-9	Coordinator exports an anonymised CSV summary for agency review.
UC-12 View Assigned Cases	Field Rescue Team	FR-7	Team member sees assigned cases sorted by priority, with location and care flags.

5.2 Use-case diagram

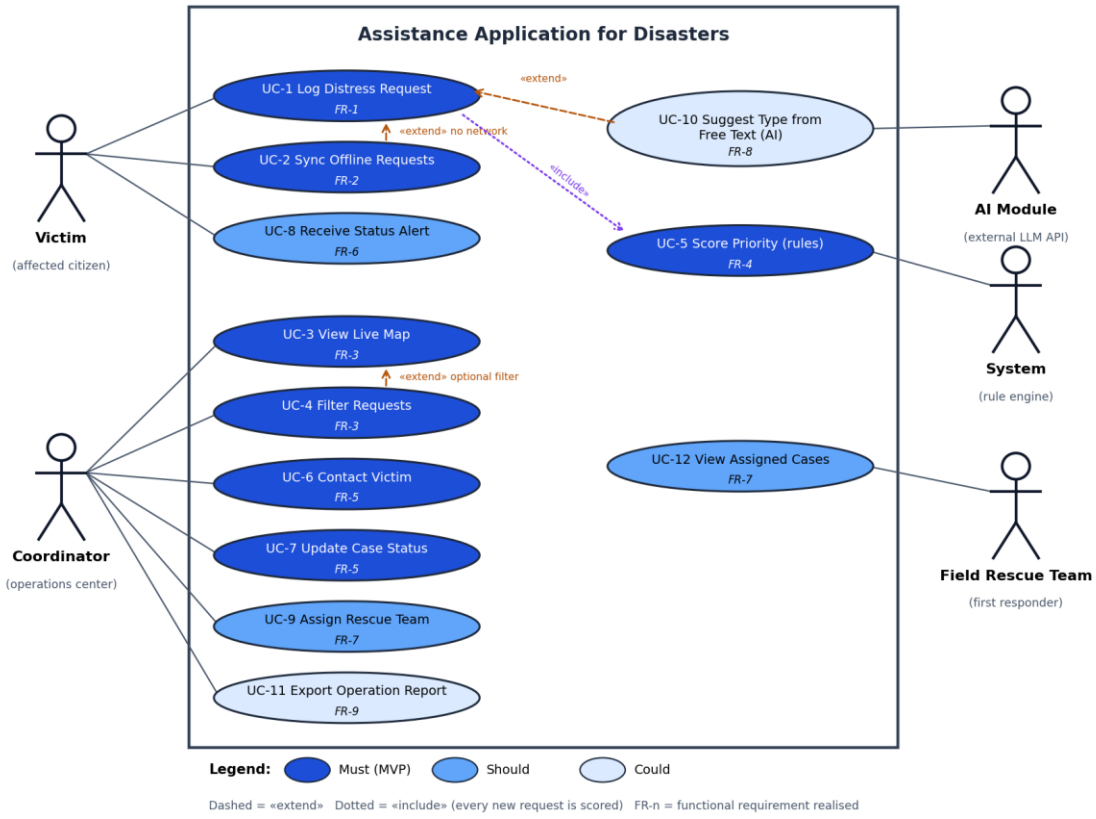


Figure 1 — Use cases by actor. Dark blue = Must (MVP), medium = Should, light = Could. Each oval shows the FR it realises.

6. Out of scope

1. No internal text chat capabilities between victims, volunteers, or rescue personnel.
2. No AI-driven medical diagnostic tools or clinical advice capabilities to answer health and treatment questions.
3. No integration to read, fetch, or summarize messages from external chat platforms or social media groups.
4. No accident / vehicle-breakdown reporting (v1.0 UC-2): a different domain with its own triage rules; possible future module.
5. No live integration with external agency systems such as Police or Ambulance (v1.0 UC-12/13): needs partner agreements; instead an anonymised CSV export is provided (FR-9).
6. No AI-based priority scoring or chatbot assistant (v1.0 UC-7): priority must be explainable and testable in a life-critical context; AI only suggests the emergency type (FR-8).
7. No payment processing, donations or subscriptions.
8. No route optimisation or turn-by-turn navigation inside the app: FR-7 only links to an external maps app.

7. Traceability matrix (Golden Thread)

Every FR is traced all the way across — problem → requirement → design → feature → test — following the single thread this course follows week by week (Ch3 → Ch4/5 → Ch6 → Ch7). Design, Feature and Test cells name the planned artefact; the Design document and the prototype will replace “planned” with links to the real files and screens.

Problem / pain	Requirement (FR)	Solution / design	Feature built	Test
P1 Victims struggle to describe locations; teams miss exact coordinates.	FR-1 Victim Request Logging (Must)	UC-1; entity Request (id, lat, lng, type, flags, phone, status); sequence: submit request	S-1 Request form	AC-1.1 – 1.5; NFR-3
P2 Unstable networks cause silent failures; requests are lost.	FR-2 Offline Capture & Sync (Must)	UC-2; local queue + client request ID; sequence: offline submit → sync	S-1 + S-2 “not sent yet” banner	AC-2.1 – 2.4; NFR-4
P3 Coordinators lack spatial context to identify high-risk clusters.	FR-3 Rescue Coordinator Live Map (Must)	Request entity + real-time map rendering job (Ch4–5); UC-3, UC-4	“Live Rescue Map” dashboard view S-3 (Ch6)	fetch_active_requests_map() unit tests (Ch7); AC-3.1 – 3.5; NFR-1, NFR-7
P4 Incoming request volume overwhelms centers; critical cases are buried.	FR-4 Priority Scoring, rule-based (Must)	UC-5; scoring rule table (§3); activity: score request	S-4 Case queue (colour + sort); pin colour on S-3	AC-4.1 – 4.5; rule-table unit tests; NFR-8
P5 Teams waste time navigating flooded zones to false or changed locations.	FR-5 Contact Victim & Case Status (Must)	UC-6, UC-7; entity StatusLog; state diagram of case status	S-4 Case detail (Call, status)	AC-5.1 – 5.4; NFR-5, NFR-7
P6 Victims cannot tell whether help is coming.	FR-6 Victim Status Notification (Should)	UC-8; notification service; sequence: status change → push	S-2 My request / status	AC-6.1 – 6.4; NFR-2
P7 Rescue teams deploy blindly.	FR-7 Assign Team & Field Case View (Should)	UC-9, UC-12; entities Team, Assignment	S-5 Assign team; S-6 My cases	AC-7.1 – 7.4; NFR-5
P8 Full forms are slow in a crisis.	FR-8 AI Suggestion from Free Text (Could)	UC-10; LLM adapter, 5 s timeout, manual fallback	S-1 “describe in one sentence” box	AC-8.1 – 8.3 (10 fixed sample sentences)
P9 Agencies need transparent logs without personal data.	FR-9 Operation Report Export (Could)	UC-11; export query on StatusLog (IDs only)	S-7 Export report	AC-9.1 – 9.3; NFR-11

Prototype note (M3): the Must screens are S-1 to S-4 — four screens that cover FR-1 … FR-5 with a clickable happy path plus at least one empty/error state (e.g. AC-3.4, AC-1.3).

8. AI usage log

If this table is empty but the prose looks generated, the milestone is capped at 50% — log every prompt that drafted a story or paragraph, honestly.

Date	Tool / model	What we asked	What AI produced	What we changed / verified	What we learned

Date	Tool / model	What we asked	What AI produced	What we changed / verified	What we learned
5 September 2026	Gemini	“Analyze recent flooding disasters in Southeast Asia (Myanmar 2025, Thailand 2025–2026) and outline key operational bottlenecks faced by rescue agencies and victims.”	A detailed overview of regional flooding impacts, highlighting issues like severed communications, uncoordinated rescue efforts, and delays in identifying high-risk victims (elderly, infants).	Synthesized the generic summary into a structured 1.1 Purpose and scope section; filtered out macro-economic statistics to focus strictly on actionable software-level pains (e.g., lack of real-time GPS coordinates and live operational visibility).	AI provided strong historical background context, but contained overly general commentary; we learned to distill raw AI background research directly into concrete “pain points” that trace directly to functional requirements.
5 October 2026	Claude	“Rewrite our v1.0 SRS to score higher on the M4 rubric for the SRS (7 sections, MoSCoW, measurable NFRs, happy + edge acceptance criteria, INVEST, Golden Thread), keeping our template.”	SRS v2.0: aligned FR/UC numbering, added MoSCoW, pains P1–P9, 11 NFRs, 37 acceptance criteria, INVEST table, a new use-case diagram and a full traceability matrix.	Kept the five original FRs, stakeholder table and pain wording. FR-6 … FR-9, all numeric targets and the scoring rules are AI-proposed and still need team review and validation with a real stakeholder before we rely on them.	AI is strong at cross-checking IDs and structure across sections; numeric targets are proposals to validate, not facts.

9. Self-check

☑ Every FR is a user story with a benefit — all 9 FRs in §3 end in a “so that” clause tied to a real pain (P1–P9).

☑ Every FR has acceptance criteria you could actually test — 37 criteria in total, each FR with at least one [Happy] and one [Edge], each a number, a state, or a yes/no.

☑ No FR is vague — no bare “fast”, “easy”, or “user-friendly” anywhere in §3 or §4.

☑ NFRs exist, including security / privacy — 11 NFRs with numbers; NFR-5 is a required access-control check, plus NFR-6 (TLS) and NFR-11 (data minimisation).

☑ Every FR traces back to a stated problem — see “Pain this traces to” under each FR, the problem register in §1.1, and §7.

☑ Scope and out-of-scope are stated — §2.1 and §6.

☑ Use-case list matches the FR list — 12 use cases, each pointing to an existing FR (diagram in §5).

☑ AI log is honest — §8 shows exactly what was asked, drafted, and what still needs team verification.

☑ Priorities are explicit — MoSCoW on every FR; Must = MVP = FR-1 … FR-5 (5 Must, 2 Should, 2 Could).

☑ INVEST gate passed — §3 INVEST table; five stories carry a noted mitigation.

Course rule: you may use AI to draft. You must log it, verify it against a real stakeholder, and be able to explain every FR in the oral exam. Work you cannot explain earns no credit.